Omii Chauhan

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Summary

Software Engineer with experience in backend development, data engineering, and machine learning orchestration. Experienced in developing scalable APIs, building automated ETL data pipelines, and implementing high-throughput multi-agent frameworks. Proficient in Python (asyncio), SQL, C++, TypeScript, Docker, and Redis, with a track record of implementing robust graph algorithms, resolving synchronization race conditions, and hardening security in open-source systems.

Education

National Institute of Technology, Warangal — **Aug 2023 – May 2027**
Bachelor of Technology in Metallurgy and Materials Science — *Warangal, India*

Telangana State Board of Intermediate Education — **Jun 2020 – May 2022**
Class XII (Higher Secondary) — 96% — *Hyderabad, India*

Central Board of Secondary Education — **Jun 2018 – May 2020**
Class X (Secondary School) — 93% — *Hyderabad, India*

Experience

LangGraph (langchain-ai) — **Jul 2026 – Present**
Open Source Contributor — *Remote*

- Patched a critical event-loop freezing bug in `ToolNode.ainvoke()` caused by silent Server-Sent Events (SSE) read timeouts in Model Context Protocol (MCP) clients (PR #8357).
- Engineered a configurable async timeout mechanism using `asyncio.wait_for` to wrap tool executions, preventing indefinite hangs.
- Resolved channel compilation behaviors to preserve Pydantic field defaults and default factories (e.g., `default_factory=lambda: ...`) when using annotated reducer functions (Issue #5225, PR #8361).
- Standardized exception handling to return structured error `ToolMessage` objects or cleanly propagate errors per configuration.

ADEN (Y Combinator) — **Feb 2026 – Present**
Core Framework Contributor — *Remote*

- Engineered a per-key asynchronous locking strategy to resolve a critical TOCTOU race condition and refactored the file-spillover pipeline to isolate corrupted states, preventing silent execution failures on resume.
- Refactored legacy synchronous tool integrations (Exa, Apollo) to native asynchronous execution using `httpx.AsyncClient`, eliminating event-loop starvation and improving tool response latency.
- Implemented the core engine for the Neutrosophic Scoring Module (RFC #7179), introducing multi-dimensional confidence logic for swarm decision-making and consensus verification.

Rajputana Vehicles Private Limited — **May 2025 – Jun 2025**
Data Analyst Intern — *Madhya Pradesh, India*

- Developed an automated ETL data pipeline using Python (Pandas) to process raw business data, eliminating 10+ hours/week of manual reporting.
- Architected customer segmentation models (K-Means) on 50k+ records, improving targeting precision by 15%, and designed Power BI EV metrics dashboards.

Technical Projects

Visual Workflow Builder | GitHub | Live Demo — **Jan 2026 – Feb 2026**
Node-based pipeline editor with backend Cycle Validation API.

- Developed a FastAPI backend using Kahn's Algorithm (Topological Sort) to perform real-time Directed Acyclic Graph (DAG) cycle detection and pipeline analysis in $O(V + E)$ complexity, ensuring cycle-free execution safety.
- Architected an extensible BaseNode component abstraction (Zustand/React Flow), reducing UI boilerplate by 40% and allowing dynamic variable parsing via regex.

AI Multi-Strategy Fund Platform | GitHub — **Nov 2025 – Dec 2025**
Autonomous agent-driven investment lifecycle and portfolio orchestration platform.

- Built an asynchronous multi-agent orchestration workflow using LangGraph and Redis queues, enabling parallel execution of independent agent allocation signals.
- Implemented hierarchical risk parity-based allocation logic to actively rebalance portfolios using historical market data, optimizing risk-adjusted returns.

FinTech AI: Credit Risk Prediction Engine | GitHub | Live Demo — **Sep 2024 – Feb 2025**
Real-time loan default prediction and batch scoring engine.

- Engineered a containerized batch processing engine (Docker) capable of scoring 10,000+ loan applications asynchronously via FastAPI workers, computing Exposure at Default (EAD) and Loss Given Default (LGD) metrics in under 5 seconds.
- Developed an end-to-end ML pipeline (Random Forest & XGBoost ensemble) to predict borrower default risk, incorporating robust feature engineering and probability calibration to improve prediction precision.

Skills

Languages: Python (asyncio), SQL, C++, TypeScript, HTML/CSS, JavaScript
Backend & Data Infrastructure: FastAPI, Docker, Redis, PostgreSQL, Snowflake, dbt, REST APIs, JSON, CORS, Linux, Git
Data Science & ML: Pandas, NumPy, Scikit-learn, XGBoost, LangGraph, K-Means Clustering, Feature Engineering
Frontend & Tools: Next.js, React, Zustand, React Flow, Power BI

Leadership & Achievements

Leadership: Head of PR & Sponsorship, Technozion & Springspree (Managed 50+ students & corporate sponsors).